

Set 13: Geographic Data Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To visualise the geographic distribution of cities on a map, explore the relationship between average temperature and population with a scatter plot, present the data in a labelled table, and combine them into an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard). The R package *maps* was used for the map outline.

Dataset

City	Population	Avg. Temperature	Elevation
City A	500,000	75	1000
City B	700,000	68	800
City C	600,000	80	1200

Program – Dataset Creation (common to all R tasks)

```
options(scipen = 999)
library(ggplot2)
library(gridExtra)
library(maps)
geo <- data.frame(
  City      = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temp  = c(75, 68, 80),
  Elevation  = c(1000, 800, 1200),
  # The dataset has no location columns; these coordinates are ASSUMED
  # (illustrative points in South India) so the cities can be placed on a map.
  Latitude  = c(11.0, 13.0, 10.0),
  Longitude = c(77.0, 80.2, 78.1)
)
write.csv(geo, "Geographic_Data.csv", row.names = FALSE) # for Tableau
```

Q1. Map Chart – Distribution of Cities

Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

R Program:

```
# Q1. Map chart - distribution of cities on a geographic map
print(geo[, c("City", "Latitude", "Longitude", "Population")], row.names = FALSE)
india <- map_data("world", region = "India")

p_map <- ggplot() +
  geom_polygon(data = india, aes(long, lat, group = group),
    fill = "#F2F2F2", colour = "grey55", linewidth = 0.3) +
  geom_point(data = geo, aes(Longitude, Latitude, size = Population, colour = Avg_Temp),
    alpha = 0.85) +
  geom_text(data = geo, aes(Longitude, Latitude,
    label = paste0(City, "\nElev. ", Elevation, " m")),
    vjust = -1.3, size = 3.3, lineheight = 0.9) +
  scale_size_area(max_size = 14, labels = scales::label_comma(), name = "Population") +
  scale_colour_gradient(low = "#FDD49E", high = "#B30000", name = "Avg. Temp (\u00B0F)") +
  coord_quickmap(xlim = c(74, 83), ylim = c(8, 15.5)) +
  labs(title = "Geographic Distribution of Cities",
    subtitle = "Bubble size = Population, colour = Avg. Temperature (locations assumed)",
    x = "Longitude", y = "Latitude") +
```

```
theme_minimal(base_size = 11) + theme(plot.title = element_text(face = "bold"))
print(p_map)
ggsave("s13_q1_map.png", p_map, width = 7.5, height = 6, dpi = 150, bg = "white")
```

Console Output:

City	Latitude	Longitude	Population
City A	11	77.0	500000
City B	13	80.2	700000
City C	10	78.1	600000

Output:

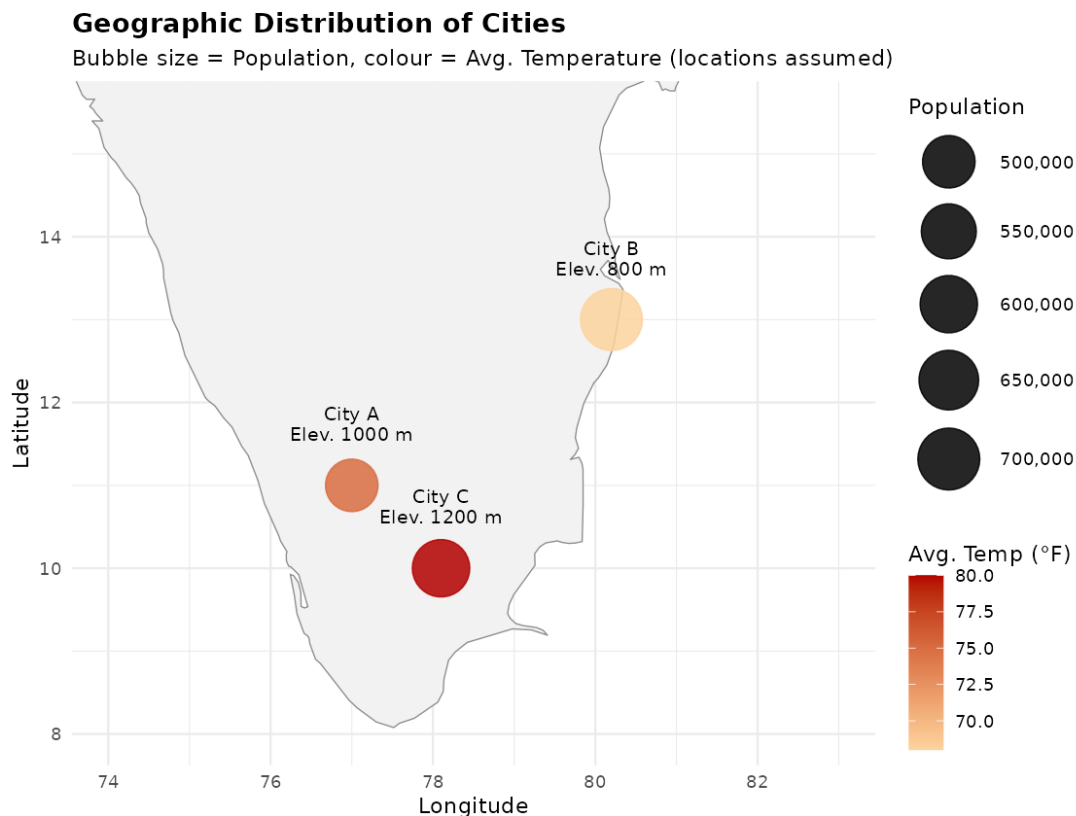


Figure 1: Cities on a map – bubble size = population, colour = average temperature

Inference:

Note: the dataset has no latitude/longitude columns, so the cities were given **assumed coordinates** (illustrative points in South India) to place them on a map. In practice, real coordinates would be added from a geocoding source.

On the map, City B is the largest bubble (700,000 people) and the coolest (68°F, lightest colour). City C is the hottest (80°F, darkest red) and has the highest elevation (1,200 m). City A is the smallest city (500,000). The labels give each city's name and elevation, and the legends explain bubble size and colour.

Q2. Scatter Plot – Average Temperature vs Population

Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

R Program:

```
# Q2. Scatter plot - Avg. Temperature vs Population
cat("Correlation (Avg_Temp vs Population):", round(cor(geo$Avg_Temp, geo$Population), 3), "\n")
cat("Correlation (Elevation vs Avg_Temp) :", round(cor(geo$Elevation, geo$Avg_Temp), 3), "\n")
cat("Correlation (Elevation vs Population):", round(cor(geo$Elevation, geo$Population), 3), "\n")

p_sc <- ggplot(geo, aes(x = Avg_Temp, y = Population)) +
  geom_point(aes(size = Elevation, colour = "#0072B2", alpha = 0.7)) +
  geom_text(aes(label = City), vjust = -1.6, size = 3.8) +
  scale_size_area(max_size = 12, name = "Elevation (m)") +
```

```
scale_x_continuous(limits = c(65, 83)) +
scale_y_continuous(limits = c(450000, 760000), labels = scales::label_comma()) +
labs(title = "Average Temperature vs Population",
     x = "Average Temperature (\u00B0F)", y = "Population") +
theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
print(p_sc)
ggsave("s13_q2_scatter.png", p_sc, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
Correlation (Avg_Temp vs Population): -0.581
Correlation (Elevation vs Avg_Temp) : 0.995
Correlation (Elevation vs Population): -0.5
```

Output:

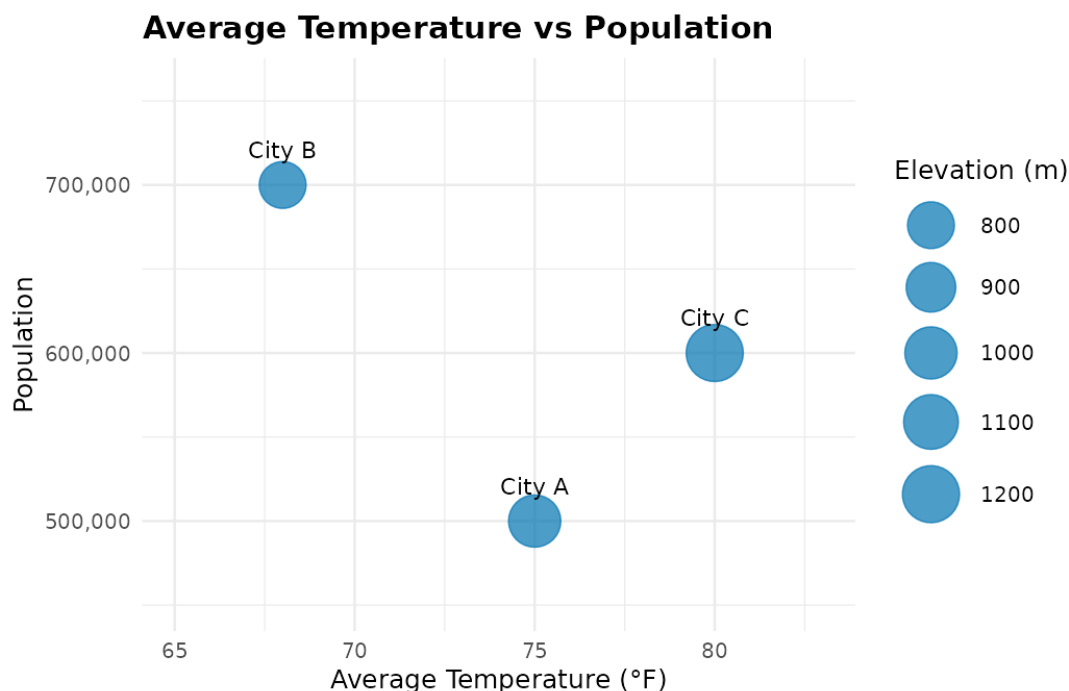


Figure 2: Average temperature vs population; bubble size = elevation

Inference:

Insights: there is a moderate **negative** relationship between temperature and population ($r = -0.58$). The coolest city (City B, 68°F) has the largest population (700,000), but the hottest city (City C, 80°F) has more people than City A (75°F), so the relationship is not consistent.

Elevation is almost perfectly correlated with temperature in this data ($r = 0.995$): higher cities are warmer, which is unusual and worth checking against the source data. With only three cities, none of these relationships can be generalised.

Q3. Table – Geographic Data for Each City

Build a table to display the geographic data for each city. Label the table elements.

R Program:

```
# Q3. Labelled table of geographic data
tbl <- geo[, c("City", "Population", "Avg_Temp", "Elevation")]
tbl$Population <- format(tbl$Population, big.mark = ",", scientific = FALSE)
names(tbl) <- c("City", "Population", "Avg. Temperature (\u00B0F)", "Elevation (m)")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontsize = 11)),
  colhead = list(fg_params = list(fontsize = 12, fontface = "bold", col = "white"),
    bg_params = list(fill = "#1F3864"))))
title <- grid::textGrob("Geographic Data by City",
  gp = grid::gpar(fontsize = 14, fontface = "bold"))
png("s13_q3_table.png", width = 1000, height = 330, res = 150)
grid.arrange(title, tg, ncol = 1, heights = c(0.25, 1))
invisible(dev.off())
```

Console Output:

```
City Population Avg. Temperature (°F) Elevation (m)
City A      500,000             75         1000
City B      700,000             68          800
City C      600,000             80         1200
```

Output:

Geographic Data by City

City	Population	Avg. Temperature (°F)	Elevation (m)
City A	500,000	75	1000
City B	700,000	68	800
City C	600,000	80	1200

Figure 3: Labelled geographic data table

Inference:

The table lists every city with labelled headers and units: Population, Avg. Temperature (°F) and Elevation (m). Total population across the three cities is 1,800,000.

Q4. Tableau Dashboard – Interactive Exploration

Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

R Program:

```
# Q4. Dashboard layout preview (Tableau dashboard built from Geographic_Data.csv)
library(patchwork)
dash <- (p_map | p_sc) / wrap_elements(full = tg) +
  plot_layout(heights = c(1.5, 0.5)) +
  plot_annotation(title = "Geographic Data Dashboard",
    theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s13_dashboard.png", dash, width = 13, height = 7.5, dpi = 130, bg = "white")
```

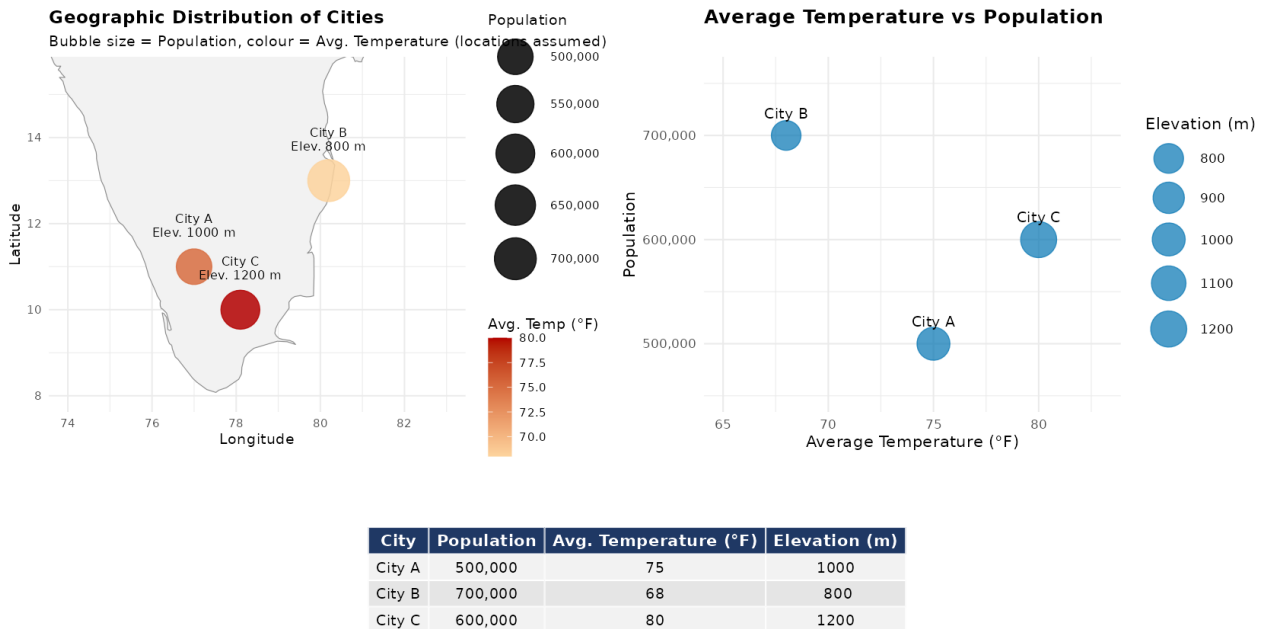
Procedure in Tableau:

- 1 Open Tableau → **Connect** → **Text file** → **Geographic_Data.csv**. It includes the Latitude and Longitude columns. Right-click Latitude → **Geographic Role** → **Latitude**, and Longitude → **Geographic Role** → **Longitude**.
- 2 **Sheet 1 – "City Map"**: double-click Longitude and Latitude to create a map, then drag City to **Detail** and **Label**. Drag Population to **Size** and Avg Temp to **Color** (sequential Orange-Red). Add Elevation to Tooltip. Title: "Geographic Distribution of Cities".
- 3 **Sheet 2 – "Temp vs Population"**: drag Avg Temp to Columns and Population to Rows. Drag City to Detail and Label, and Elevation to Size. Optionally add **Analytics** → **Trend Line**.
- 4 **Sheet 3 – "City Table"**: drag City to Rows and Measure Names to Columns. Put Measure Values (Population, Avg Temp, Elevation) on Text. Rename the headers with units.
- 5 From the menu choose **Dashboard** → **New Dashboard**. Set **Size** → **Automatic** (or Fixed 1200 × 800).
- 6 Drag the sheets onto the canvas: City Map (left), Temp vs Population (right), City Table (bottom). Add a **Text** object at the top for the dashboard title.
- 7 On each chart, click the sheet's drop-down and select **Use as Filter**. Clicking a bar or slice then filters the other views.
- 8 Go to **Dashboard** → **Actions** → **Add Action** → **Highlight** (Run on: Hover) so related marks highlight across all views.
- 9 Right-click a filter → **Apply to Worksheets** → **All Using This Data Source** (Geographic_Data) so one filter controls every view.

- 10 Save as a packaged workbook (**File** → **Save As** → **.twbx**) or publish to Tableau Public.
- 11 On the map, choose **Use as Filter**. Clicking a city then filters the scatter plot and table to that city.

Dashboard Layout:

Geographic Data Dashboard



Reference layout of the dashboard (rendered in R). The interactive version is built in Tableau using the procedure above.

Inference:

Selecting a city on the map highlights it in the scatter plot and filters the table, so its location, climate and population can be compared at a glance.

Result

The geographic data was visualised with a map chart (using assumed coordinates), a scatter plot and a labelled table in R, and combined into an interactive Tableau dashboard. City B is the largest and coolest city, and City C is the hottest and highest. Temperature and population show a moderate negative relationship ($r = -0.58$) in this small sample.